BLOCKCHAIN-SECURED SIM SWAP FRAUD DETECTION SYSTEM: AN END-TO-END MULTIDISCIPLINARY FRAMEWORK

PROJECT REVIEW I DOCUMENTATION

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ABSTRACT

SIM swap fraud presents a critical vulnerability in telecommunication networks, enabling attackers to bypass Multi-Factor Authentication (MFA) and compromise sensitive user data. Current mitigation strategies rely heavily on manual verification or isolated machine learning models that lack interpretability and are susceptible to insider threats. This project proposes a multidisciplinary framework combining hardware-level signal extraction, Explainable Artificial Intelligence (XAI), and distributed ledger technology. The architecture is divided among three engineering disciplines: ECE handles edge-device telemetry and signal integrity, CSE implements anomaly detection via machine learning (XGBoost) and SHAP-based explainability, and IT orchestrates a cybersecurity risk engine integrated with an Ethereum-based blockchain for immutable auditing. By logging both the decision output and the human-readable algorithmic reasoning on-chain, this system provides a tamper-evident, automated solution to SIM swap fraud detection.

CHAPTER 1: INTRODUCTION

1.1 Problem Statement

SIM swap fraud is an unauthorized porting of a telecommunications subscriber's number to a new hardware Subscriber Identity Module (SIM). This exploit is primarily executed through social engineering tactics directed at telecom customer service representatives or through direct insider collusion. Upon a successful swap, the malicious actor gains the ability to intercept incoming calls and Short Message Service (SMS) data. Consequently, this enables the bypass of Multi-Factor Authentication (MFA) protocols, leading to unauthorized access to banking institutions, cryptocurrency exchanges, and personal identity records.

Conventional rule-based security measures, such as identity verification questions, are increasingly inadequate. Furthermore, reliance on human intervention introduces a single point of failure within the telecommunications operational pipeline.

1.2 Proposed System Objectives

The primary objective of this project is to develop a decentralized, intelligent, and verifiable verification system for SIM replacement requests. The objectives include:

- Automating the anomaly detection process using historical user behavior and real-time telecom telemetry.
- Providing deterministic governance through a hybrid cybersecurity risk engine that validates machine learning predictions against hard-coded security rules.
- Implementing Explainable AI (XAI) to ensure the system's decisions are transparent and interpretable by human operators.
- Securing the decision-making pipeline utilizing blockchain technology to prevent unauthorized post-incident modifications of the audit logs.

CHAPTER 2: LITERATURE REVIEW

2.1 Current Methodologies in Fraud Detection

The literature indicates a significant transition towards utilizing Artificial Intelligence for telecommunications security. Recent studies demonstrate the efficacy of utilizing models such as Random Forest and Isolation Forests for evaluating Call Detail Records (CDRs) and identifying rapid SIM state changes. While these models yield high detection accuracy, they operate inherently as "black boxes," providing binary classifications without contextual reasoning.

2.2 Blockchain Applications in Auditing

To mitigate insider threats, distributed ledger technology has been proposed for security auditing. Existing frameworks commonly map machine learning outputs directly to Ethereum-compatible ledgers. However, the application of blockchain in current literature predominantly targets e-commerce transaction anomalies or spam caller identification. Furthermore, these architectures lack an intermediate validation layer, meaning erroneous machine learning predictions are recorded directly to the ledger.

2.3 The Gap in Explainable AI (XAI) Integration

Regulatory frameworks increasingly mandate a "Right to Explanation" for automated decision-making systems. While techniques like SHapley Additive exPlanations (SHAP) are prevalent in financial sectors, their integration into blockchain-secured telecom networks remains theoretical. Existing systems log the final algorithmic decision but fail to cryptographically secure the reasoning parameters underlying that decision.

2.4 Novelty and Contribution

The proposed framework addresses the identified literature gaps by introducing an intermediate Cybersecurity Risk Engine and on-chain XAI. Table 2.1 summarizes the distinctions between the proposed system and existing methodologies.

Table 2.1: Comparison of Existing Literature and Proposed System

Feature	Existing Literature	Proposed Architecture
Application Domain	Spam call detection, financial transaction fraud.	Telecommunications SIM swap fraud detection.
Decision Routing	Direct logging of ML binary output (Fraud/Not Fraud).	Three-tier output (Approve, Review, Block) governed by a Risk Engine.
Explainability	Not recorded on the ledger.	SHAP vectors generated and hashed on-chain alongside the decision.
System Scale	Offline evaluation via synthetic datasets.	Live multi-tier prototype integrating edge hardware, cloud API, and blockchain.

CHAPTER 3: SYSTEM ARCHITECTURE

3.1 Multidisciplinary Integration

The architecture is designed to enforce a strict separation of concerns across three distinct engineering disciplines, ensuring independent validation at the hardware, analytical, and governance levels.

3.2 Departmental Modules

3.2.1 Electronics & Communication Engineering (ECE) - Hardware Edge

The ECE module focuses on physical data acquisition and signal integrity. Using GSM modules (e.g., SIM800L) and microcontroller gateways, this layer emulates the physical SIM request environment. The primary responsibility is extracting unaltered hardware-level telemetry, including International Mobile Equipment Identity (IMEI) data, cell tower identifiers, and physical signal metrics, before transmission to the cloud architecture.

3.2.2 Computer Science & Engineering (CSE) - Data Science & AI

The CSE module is responsible for algorithmic anomaly detection. This involves training supervised learning models (e.g., XGBoost) on behavioral telecom datasets. Additionally, this layer incorporates the SHAP library to process the model's outputs, dynamically generating text-based reasoning vectors that explicitly state the statistical features contributing to the anomaly score.

3.2.3 Information Technology (IT) - Risk Engine and Blockchain Integration

The IT module orchestrates the system's governance and security infrastructure. A high-throughput backend application program interface (API) acts as the Cybersecurity Risk Engine, ingesting the CSE module's anomaly scores and cross-referencing them against relational databases of blacklisted parameters. Based on these inputs, it calculates a final decision (Auto-Approve, Manual Review, or Auto-Block). Subsequently, a Solidity-based smart contract deployed on an Ethereum test network permanently logs a cryptographic hash of both the final decision and the corresponding SHAP explanation, creating an immutable audit trail accessible via an administrative dashboard.